SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 - SOFTWARE TESTING

EXPERIMENT 15

Comprehensive Test Plan – Inventory Management System

Domain: Inventory Management System

1. AIM

To develop a comprehensive Test Plan for an Inventory Management System covering test scope, test design techniques, test execution strategy, and test completion criteria.

2. PROBLEM STATEMENT

An Inventory Management System manages items, stock quantities, suppliers, purchase/issue transactions, reorder levels, alerts, and reports. The testing team must verify stock accuracy, transaction processing, access control, and reporting.

3. SOFTWARE REQUIREMENT SPECIFICATION (SRS)

S.No.	Requirement
1	The system must allow valid item creation and updates.
2	Stock receipt and issue quantities must be updated correctly.
3	Supplier records must be maintained correctly.
4	Low-stock alerts must be generated at configured levels.
5	Inventory reports must be accurate.
6	Users must have only permitted inventory operations.
7	Invalid quantities and duplicate transactions must be rejected.

4. TEST PLAN / TEST STRATEGY

Test Area	Approach
Item Management	Add, edit, search and deactivate inventory items.
Stock Management	Receive, issue, adjust and verify stock quantities.
Supplier Management	Validate supplier records and purchase orders.
Low-stock Alerts	Verify alerts at configured reorder levels.
Reports	Verify stock and transaction reports.
User Access	Verify role-based permissions for inventory operations.

5. TEST DESIGN TECHNIQUES

Technique	Purpose
Equivalence Partitioning	Test valid and invalid item, quantity and supplier classes.
Boundary Value Analysis	Test reorder levels, minimum quantities and maximum limits.
Decision Table Testing	Verify stock, supplier, approval and reorder combinations.
State Transition Testing	Verify Available, Reserved, Issued, Low Stock and Out of Stock states.
Pairwise Testing	Cover important combinations of inventory parameters.

6. TEST EXECUTION STRATEGY

1. Review approved requirements and prepare test data.
2. Execute smoke testing on each test build.
3. Execute positive and negative functional test cases.
4. Execute integration and system testing.
5. Record actual results and log defects.
6. Retest fixed defects and perform regression testing.
7. Prepare the final test summary.

7. TEST ENVIRONMENT

Item	Specification
Application	Inventory Management System
Operating System	Windows 10/11
Browser	Google Chrome / Microsoft Edge
Database	Controlled test database
Network	Stable test network
Test Data	Valid, invalid, boundary, duplicate and missing data

8. ENTRY CRITERIA

Approved requirements are available; test build is deployed; test environment is ready; test data is prepared; major dependencies are available.

9. TEST COMPLETION / EXIT CRITERIA

All planned critical tests are executed; no open Critical defects remain; High-severity defects are resolved or formally accepted; failed tests are analyzed; regression testing is completed; and the final test summary is prepared.

10. DEFECT REPORTING

Every defect should contain a defect ID, title, linked test case, severity, priority, environment, preconditions, steps to reproduce, expected result, actual result, impact, suggested fix, and status.

11. RESULT

The comprehensive test plan for the Inventory Management System was prepared with defined scope, test design techniques, execution strategy, defect reporting method, and completion criteria.